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The fixed- σ slices of the residue-dependent solvability obstruction o_ν : additivity, B^2 -killing, and the torsor dichotomy

Corresponds to work-in-progress commit COMMITHASH (github.com/scotmacbeth/work-in-progress).

Rick — here are the fixed- σ slices you asked for (your email of 2026-09-29, targets (a) and (b)), presented against your two read-points. The underlying object is the obstruction we settled on 09-29; this note slices it by the circle action σ and reports the per-slice tables. Trust grades are stated inline and kept honest: the *slice tables* are **computed** (verification), the *main theorem* they instantiate is **proved**, and the general *torsor* claim remains **computed** (not yet proved in general).

1 Setup and what is already proved

Fix a good triplet (ν, μ, σ) for a skew-brace extension $0 \rightarrow D \rightarrow G \rightarrow H \rightarrow 0$ with D an ideal: $\nu : (H, \circ) \rightarrow \text{Aut}(D, +)$ a homomorphism (the residue channel), $\mu : (H, +) \rightarrow \text{Aut}(D, +)$ (trivial here, as $(G, +)$ is abelian), and $\sigma : (H, \circ) \rightarrow \text{Aut}(D, +)$ an anti-homomorphism (the circle action). Reading Rathee–Yadav (3.10) as an affine-linear equation in the multiplicative datum for fixed β gives two $\mathbb{Z}$ -linear operators $P, Q : C^2 \rightarrow C^3$; with $T(\bar{\tau})(h_1, h_2) = \nu_{h_1 \circ h_2}(\bar{\tau}(h_1, h_2))$ and $A := P \circ T$ restricted to $Z_\circ^2 := Z_{\text{Gp}}^2((H, \circ), D; \sigma)$, the obstruction is

$$o_\nu : H_+^2 \longrightarrow C^3/A(Z_\circ^2(\sigma)), \quad o_\nu([\beta]) = [Q\beta].$$

Proved (09-29). In the trivial-brace-kernel (RY) regime, for general H : o_ν is a well-defined group homomorphism that kills B_+^2 , and

$\text{im } \varphi^+ = \ker o_\nu$

— an additive class $[\beta]$ extends to a full skew-brace cocycle inducing (ν, μ, σ) iff $o_\nu([\beta]) = 0$. This is **proved** (structural; the linchpin identity $P(T d_\circ \theta_1) = Q(d_+ \theta)$ is a formal consequence of $B^2 \subseteq Z^2$, exhaustively machine-checked; $\ker o_\nu = \text{im } \varphi^+$ cross-validated against an independent regular-subgroup enumeration). Full statement and proof: `proofs/2026-09-29-solvability-obstruction-o-nu.tex`, Theorem 1.

The fixed- σ slice framing (your Remark 4). The target $A(Z_\circ^2(\sigma))$ depends on σ , while the domain H_+^2 (computed with μ trivial) does not. So it is natural to slice: fix σ , vary the admissible residue ν , and read $o_\nu \upharpoonright_\sigma$ as a map $H_+^2 \rightarrow C^3/A(Z_\circ^2(\sigma))$. Your two read-points then become per-slice questions, answered below.

2 Read-point (a): each slice is additive and kills B_+^2

Claim (a). For every fixed σ and admissible ν , the slice $o_\nu \upharpoonright_\sigma$ is an *additive* map $H_+^2 \rightarrow C^3/A(Z_\circ^2(\sigma))$ that *kills* B_+^2 ; hence $\ker o_\nu \upharpoonright_\sigma = \text{im } \varphi^+$ is a based *subgroup* of H_+^2 . This is the slice restriction of Theorem 1 (**proved**); the tables below are the **computed** verification.

I verified, per slice, (i) $o_\nu(\beta_1 + \beta_2) \equiv o_\nu(\beta_1) + o_\nu(\beta_2) \pmod{A}$ over *all* pairs $\beta_1, \beta_2 \in Z_+^2$, and (ii) $o_\nu(d_+ \theta) \equiv 0$ over *all* additive coboundaries $d_+ \theta$. Both pass with no exceptions (`scratch/2026-09-30-o-nu-fix`).

family	slice	residue	additivity	B^2 -killing	$ \ker / H_+^2 $	basepoint
$ G =8$ $H=\mathbb{Z}/2, D=\mathbb{Z}/4$	$\sigma = \text{id}$	$\nu = \text{id}$	16/16	4/4	2/2	present
	$\sigma = \text{id}$	$\nu = -1$	16/16	4/4	2/2	present
	$\sigma = -1$	$\nu = \text{id}$	16/16	4/4	1/2	present
	$\sigma = -1$	$\nu = -1$	16/16	4/4	2/2	present
$ G =9$ $H=\mathbb{Z}/3, D=\mathbb{Z}/3$	$\sigma = \text{id}$	$\nu = \text{id}$	81/81	9/9	3/3	present
	$\sigma = \times 2$	$\nu = \text{id}$	81/81	9/9	1/3	present

Table 1: Trivial-kernel (RY) slices. Additivity checked over all $|Z_+^2|^2$ pairs; B^2 -killing over all $|B_+^2|$ coboundaries. Every slice is a homomorphism killing B_+^2 ; the kernel is a subgroup and always contains the origin (basepoint present). **computed**.

Two remarks tying this to the closed forms. First, the domain size is σ -independent ($|H_+^2| = 2$ at $|G| = 8$, $= 3$ at $|G| = 9$), while the kernel *size* moves with the slice: at $|G| = 8$, $\sigma = -1$, $\nu = \text{id}$ the kernel drops to $\{[\beta] = 0\}$ (only the split class survives), recovering “ $\nu = \text{id}$ forces $[\beta] = 0$ ”. Second, on the $H = \mathbb{Z}/2$ slices the abstract kernel agrees with the closed form

$$o_\nu(b) = 0 \iff (1 + \nu)b \in 2D^\sigma,$$

verified as “MATCHES” on every row — so the $|G| = 8$ condition $(1 + \nu)b \in 2D^\sigma$ falls out of the slice exactly as the per- σ membership test.

3 Read-point (b): the torsor dichotomy

Claim (b). A *non-trivial-brace* kernel makes $o_\nu \upharpoonright_\sigma$ *affine* rather than linear: the realising equation acquires a constant offset c , so the realisable locus $\text{im } \varphi^+$ is a *coset* of $\ker o_\nu$ — a torsor. When that offset is nonzero the coset misses the origin: *no split* ($[\beta] = 0$) *brace exists*, i.e. the torsor has **no basepoint**. This is **computed** on the $|G| = 16$ witness family and **open** for a general proof.

The two per- σ conditions. In the trivial-kernel regime (§2) the per- σ condition is the *linear/subgroup* membership $(1 + \nu)b \in 2D^\sigma$; the origin always solves it, so the basepoint is always present. In the non-trivial-kernel family $D = \mathbb{Z}/8$, $\lambda_x^D = \times 5^x$, $L = \{1, 5\} \subsetneq \text{Aut}(D)$, the residue re-enters through λ_d on the complement and the per- σ condition becomes the *affine* three-way lock

$$4b \equiv 2(\nu\sigma - 1) \pmod{8}, \quad \text{offset } c := 2(\nu\sigma - 1) \pmod{8},$$

constant in b . The realisable b form the coset $\{b : 4b = c\}$ of $\ker(\times 4) = \{0, 2, 4, 6\}$, and

$$\text{basepoint present} \iff 0 \in \{b : 4b = c\} \iff c = 0 \iff \nu\sigma \equiv 1 \pmod{4}.$$

I verified both the affine lock and this basepoint dichotomy on all 24 genuine order-16 braces with this kernel, sliced by σ .

The dichotomy is crisp: whenever the internal datum $\nu\sigma$ leaves the socle-like subgroup ($\nu\sigma \not\equiv 1 \pmod{4}$), the offset $c = 4 \neq 0$ and $\text{im } \varphi^+$ is a torsor over $\ker o_\nu$ missing the origin — $[\beta] = 0$ is *forbidden*, the split additive brace does not exist for that (ν, σ) . This is the honest replacement of the old “internal-replacement coordinate” story, and the affine offset c is the intrinsic obstruction to a basepoint.

slice	residue	realisable b	offset $c = 2(\nu\sigma - 1)$	$\nu\sigma \bmod 4$	basepoint
$\sigma = \times 1$	$\nu = \times 1$	$\{0\}$	0	1	present
$\sigma = \times 1$	$\nu = \times 3$	$\{1\}$	4	3	absent
$\sigma = \times 1$	$\nu = \times 5$	$\{0\}$	0	1	present
$\sigma = \times 1$	$\nu = \times 7$	$\{1\}$	4	3	absent
$\sigma = \times 3$	$\nu = \times 3$	$\{0\}$	0	1	present
$\sigma = \times 3$	$\nu = \times 7$	$\{0\}$	0	1	present
$\sigma = \times 5$	$\nu = \times 1$	$\{0\}$	0	1	present
$\sigma = \times 5$	$\nu = \times 3$	$\{1\}$	4	3	absent
$\sigma = \times 5$	$\nu = \times 5$	$\{0\}$	0	1	present
$\sigma = \times 5$	$\nu = \times 7$	$\{1\}$	4	3	absent
$\sigma = \times 7$	$\nu = \times 3$	$\{0\}$	0	1	present
$\sigma = \times 7$	$\nu = \times 7$	$\{0\}$	0	1	present

Table 2: Non-trivial-kernel ($|G| = 16$, $D = \mathbb{Z}/8$, $L = \{1, 5\}$) slices. In each slice the affine lock $4b \equiv c$ holds on every realising brace, and the basepoint is present iff $c = 0$ iff $\nu\sigma \equiv 1 \pmod{4}$. The rows with $\nu\sigma \equiv 3$ ($c = 4$) realise *only* $[\beta] \neq 0$: a torsor with no basepoint. **computed**.

Grade, honestly. The affine/torsor structure and the basepoint dichotomy are **computed** on this family and match every genuine brace; a general- H , general-kernel proof that $\text{im } \varphi^+$ is an o_ν -affine torsor (with c identified intrinsically as the class of the λ_d -complement cochain) is the natural next target and is **open**. Nothing in the slice computation contradicts the 09-29 proof file — the linear-slice tables of §2 and the affine-slice table of §3 reproduce its $|G| = 8$ and $|G| = 16$ conditions exactly.

Verification script: `scratch/2026-09-30-o-nu-fixed-sigma-slices.py` (reuses the four 09-29 scripts). All slices: additivity and B^2 -killing pass with no exceptions; the $|G| = 16$ affine lock and basepoint dichotomy pass on all 24 genuine braces. Happy to co-develop the general torsor proof if you want to take the offset-as-a-class route.